

The Price of Linearity: Carleman–QSVT, Variational, and Analog Routes to Quantum Simulation of Nonlinear Partial Differential Equations

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A quantum computer evolves states linearly, so a nonlinear partial differential equation must first be embedded in a linear problem. We quantify what that embedding costs, and what happens if one declines to pay it. Taking the viscous Burgers and Korteweg–de Vries (KdV) equations on a periodic domain as a common target, we compare three routes end to end. Route 1, Carleman linearization followed by quantum singular value transformation (QSVT), we implement at circuit level and verify against classical reference solutions to numerical precision; we then derive its complete error budget and find two obstructions that gate-level demonstrations do not expose. The Carleman series has a provable convergence regime only for Reynolds number $\text{Re} < \pi/2 \approx 1.57$, and the coherent-permutation block encoding carries a subnormalization $\alpha \simeq n_x^{N_C-1} \lambda_{\text{eff}}$ that erases the nominal exponential compression; with a sparse arithmetic oracle and a history-state dilation, the assembled sequential depth is $\tilde{O}(\kappa_V T N_C^3 \text{Re}^{-1} \epsilon^{-1})$, which at $\text{Re} = 1$, $\epsilon = 10^{-3}$ evaluates to 1.3×10^{12} logical two-qubit gates against 4.4×10^6 classical floating-point operations for the same instance. Route 2 declines the embedding: we amplitude-encode the field directly and propagate it by quantum natural-gradient descent, measuring every required overlap with circuits of at most 351 gates and one ancilla. This reproduces both equations to relative L^2 error 2×10^{-8} – 4×10^{-6} in ideal simulation and 0.07–0.20 under an IBM device calibration snapshot with shot noise, and its resources grow polylogarithmically in grid size, $\Theta(\log^2 n_x)$ parameters and $\mathcal{O}(\text{poly}(\log n_x))$ depth. Route 3 declines the digital circuit entirely: a digital degenerate cavity laser realizes Burgers dynamics as the continuum limit of a Sakaguchi–Kuramoto phase-oscillator array, with effective viscosity set by the real part of a programmable optical coupling. Underpinning all three, we prove that the skew-symmetric convective closure is what makes the required norm bounds hold exactly at any resolution, and that forward Euler provably injects energy into KdV on every step whereas implicit midpoint does not. The obstruction to advantage is localized, not diffuse: in one dimension the linear-embedding route buys exponential memory compression and nothing else.

POPULAR SUMMARY

Quantum computers are built to do one thing extremely well: linear algebra. Every quantum gate is a matrix, and the machine’s state evolves by matrix multiplication. That is a problem for fluid dynamics, because fluids are nonlinear — a wave’s own speed depends on its own height, which is why ocean waves steepen and break. To simulate such a flow on a quantum computer, one must first rewrite the nonlinear problem as a much larger linear one, and only then hand it to the machine.

This paper asks what that rewriting actually costs, using two textbook fluid equations as test cases. We build the standard rewriting — Carleman linearization — as a real quantum circuit, check it against classical calculations, and then work out how it scales. The answer is discouraging in a specific and useful way. The rewriting is only mathematically guaranteed to work when the flow is extremely gentle, below a Reynolds number of about 1.6, which excludes essentially every flow anyone wants to simulate; and even in that gentle regime the circuit needs roughly a trillion operations, where a laptop needs

a few million.

We then show two ways to avoid paying that price. The first keeps the nonlinearity outside the quantum circuit: a small quantum circuit stores the fluid’s shape, and a classical optimizer nudges it forward in time. This runs on today’s noisy hardware — our circuits are under 351 gates, and we test them against a real IBM processor’s calibration data. The second abandons digital circuits altogether and uses a laser: an array of coupled laser spots, whose phases obey the same equation the fluid does.

The useful result is not that one route wins, but that we can now say exactly where the cost sits. Three of the obstructions we identify are engineering problems with identifiable fixes. Only one — the convergence limit of the linearization itself — is fundamental.

I. INTRODUCTION

Quantum computers evolve states linearly. A nonlinear partial differential equation (PDE) therefore cannot

be handed to one directly: it must first be *embedded* in a linear problem, and only the embedded problem given to a quantum subroutine. Every proposed quantum algorithm for nonlinear PDEs pays for that embedding somewhere, and the central question — rarely answered in full — is how much.

We take as a common target two canonical nonlinear PDEs on the periodic spatial domain $x \in [0, 1]$,

$$\partial_t u + u \partial_x u = \nu \partial_{xx} u, \quad (\text{viscous Burgers}) \quad (1)$$

$$\partial_t u + u \partial_x u + \delta \partial_{xxx} u = 0, \quad (\text{KdV}) \quad (2)$$

with $\nu > 0$ the viscosity and $\delta > 0$ the dispersion coefficient. The pair is chosen deliberately. Burgers is parabolic and dissipative, and its exact solution obeys a maximum principle, $\max_x |u(x, t)| \leq \max_x |u(x, 0)|$; the argument is elementary, since at an interior maximum $\partial_x u = 0$ and hence $\partial_t u = \nu \partial_{xx} u \leq 0$ there. KdV is dispersive and conservative: neither term dissipates, and $\|u(t)\|_2 = \|u(0)\|_2$ exactly, because both the dispersive and the convective term do zero net work (Sec. II B). Burgers is also, as we discuss in Sec. IV G, linearizable in closed form by the Cole–Hopf substitution, which makes it a benchmark that can demonstrate that circuits compile but cannot by itself demonstrate that a method handles nonlinearity. KdV admits no such substitution, and is therefore the control experiment; we carry both throughout.

We compare three routes to simulating these equations, and the comparison is the paper.

Route 1 — pay for the embedding.:

Carleman linearization lifts the discretized quadratic ordinary differential equation (ODE) into a truncated *linear* system, which a quantum linear solver based on the quantum singular value transformation (QSVT) [7] inverts at each implicit timestep [1, 3]. We construct this at circuit level for both PDEs, with two different block encodings, and verify it (Sec. III). We then derive its complete error budget and asymptotic cost (Sec. IV).

Route 2 — keep the nonlinearity classical.:

The field is amplitude-encoded directly into a shallow parametrized circuit and stepped forward in time by quantum natural-gradient descent toward a physically motivated target, following the McLachlan variational principle [8–10]. No linear embedding is constructed at all (Sec. V).

Route 3 — make the hardware nonlinear.:

A digital degenerate cavity laser (DDCL) realizes an array of coupled phase oscillators whose continuum limit *is* the viscous Burgers equation, with the effective viscosity set by the real part of a programmable complex optical coupling (Sec. VI).

Three contributions distinguish this work from a survey of the alternatives.

TABLE I. Notation used throughout. Where the underlying literature conflicts, we adopt the convention in the right-hand column.

Symbol	Meaning
n_x	number of spatial grid points
n	qubits in the field register, $n_x = 2^n$
N_C	Carleman truncation order
$\text{Re} = UL/\nu$	Reynolds number ($U = \ u_0\ _\infty$, $L = 1$)
Re_Δ	cell Reynolds number, $U\Delta x/\nu$
ϵ	target relative error
α	block-encoding subnormalization
κ_Q	effective condition number, $\alpha/\sigma_{\min}$
κ_V	eigenvector condition number of A
P	variational ansatz parameter count
m	number of timesteps

First, all three routes are held to the same physical standard. We prove (Sec. II B) that the skew-symmetric convective closure is what makes the required L^2 bounds hold *exactly and at any resolution*, and we prove (Sec. II C) that forward Euler injects energy into KdV’s conservative ODE on every step regardless of stepsize, whereas implicit midpoint conserves it exactly. These are not incidental numerical hygiene: without the first, a discretization violates the very bound that defines the target physics, and without the second, a variational propagator inherits a systematic drift that no amount of circuit fidelity can remove.

Second, we separate *measured* from *projected* performance throughout, and label every number accordingly. Sections III and V report circuits we built, transpiled, and ran; Sec. IV reports asymptotics we derived and calibrated against published gate-level data [1]. Conflating the two is, in our reading, the main way resource estimates for this problem class become misleading.

Third, we localize the obstruction. Of the five cost drivers in Route 1, the qubit count is a clean exponential win that nothing erodes; the block encoding and the sequential time-stepping are large but fixable; the readout is a hard information-theoretic barrier only if the full field is wanted; and exactly one — the Carleman convergence radius — is fundamental within the framework. That decomposition, rather than a single headline number, is what we regard as the useful output.

Figure 1 shows the three routes as one pipeline, with the stage at which each declines to continue down the linear-embedding chain, and where each one’s cost accumulates. Table I fixes notation across the three routes, which are drawn from formulations that use conflicting symbols in the literature.

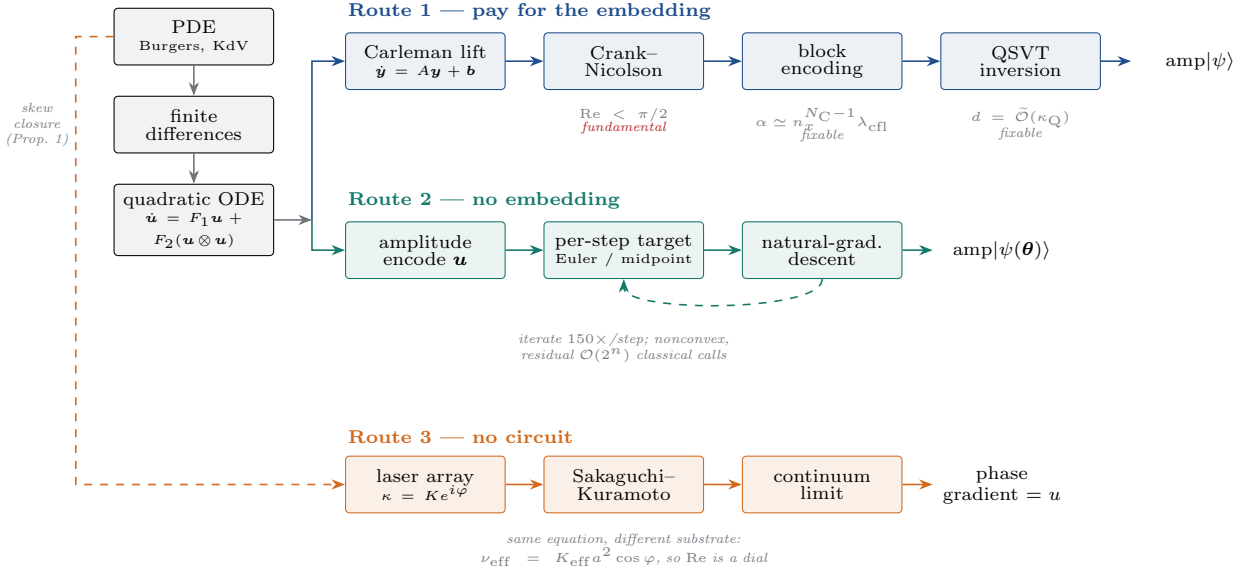


FIG. 1. The three routes as one pipeline. All share the discretization and the exactly energy-conserving convective closure (Proposition 1), then diverge at the point where each decides how to reconcile a nonlinear equation with linear hardware. Route 1 (blue) completes the linear-embedding chain and pays for it at every stage; annotations mark where the cost accrues and whether it is fixable, with the Carleman convergence threshold $\text{Re} < \pi/2$ the one obstruction that is not. Route 2 (teal) stops at the amplitude encoding and moves the nonlinearity into a classical optimizer, trading circuit depth for a nonconvex problem solved afresh each step. Route 3 (orange) never digitizes: a coupled laser array whose continuum limit is Burgers, in which the Reynolds number that obstructs Route 1 becomes a tunable coupling phase.

II. THE SHARED PHYSICAL PROBLEM

All three routes describe the same physics, and Routes 1 and 2 share one discretization. We set it up here, and prove the two structural facts that everything downstream depends on.

A. Discretization

Both PDEs are discretized on n_x periodic grid points $x_i = i/n_x$, spacing $\Delta x = 1/n_x$, with central finite differences [17]. Writing $u_i \approx u(x_i, t)$, the first, second, and third derivatives are the circulant matrices

$$(D_1)_{i,i\pm 1} = \pm \frac{1}{2\Delta x}, \quad (3)$$

$$(D_2)_{i,i\pm 1} = \frac{1}{\Delta x^2}, \quad (D_2)_{ii} = -\frac{2}{\Delta x^2}, \quad (4)$$

$$(D_3)_{i,i\pm 1} = \mp \frac{1}{\Delta x^3}, \quad (D_3)_{i,i\pm 2} = \pm \frac{1}{2\Delta x^3}, \quad (5)$$

all with periodic wraparound. Both equations then take the quadratic form

$$\dot{\mathbf{u}} = F_1 \mathbf{u} + F_2(\mathbf{u} \otimes \mathbf{u}), \quad (6)$$

with $F_1 = \nu D_2$ for Burgers, $F_1 = -\delta D_3$ for KdV, and F_2 a fixed bilinear operator representing $-u \partial_x u$ acting on the length- n_x^2 vector $\mathbf{u} \otimes \mathbf{u}$.

B. The convective closure must be chosen with care

The naive central-difference convective term, $(F_2^{\text{naive}}(\mathbf{u} \otimes \mathbf{u}))_i = -\frac{1}{2\Delta x} u_i(u_{i+1} - u_{i-1})$, does not satisfy $\mathbf{u}^\top F_2^{\text{naive}}(\mathbf{u} \otimes \mathbf{u}) = 0$ for generic $\mathbf{u}$ — only for specially symmetric inputs, by coincidence. Since $\frac{d}{dt} \|\mathbf{u}\|_2^2 = 2\mathbf{u}^\top F_1 \mathbf{u} + 2\mathbf{u}^\top F_2(\mathbf{u} \otimes \mathbf{u})$ and $\mathbf{u}^\top F_1 \mathbf{u} \leq 0$ for Burgers' dissipative F_1 , the sign of the derivative is guaranteed only if the second term vanishes identically; otherwise nothing prevents the discrete norm from growing, in direct violation of the maximum principle. We therefore use the skew-symmetric (energy-conserving) closure

$$F_2(\mathbf{u} \otimes \mathbf{u})_i = -\frac{1}{3} \left[u_i (D_1 \mathbf{u})_i + (D_1(\mathbf{u} \odot \mathbf{u}))_i \right], \quad (7)$$

with $\odot$ the elementwise product — half the naive form plus half its conservative flux counterpart, the standard structure of energy-conserving treatments of nonlinear advection [14, 15].

Proposition 1. $\mathbf{u}^\top F_2(\mathbf{u} \otimes \mathbf{u}) = 0$ identically, for every $\mathbf{u} \in \mathbb{R}^{n_x}$.

Proof. D_1 is antisymmetric: $(D_1)_{i,i+1} = \frac{1}{2\Delta x} = -(D_1)_{i+1,i}$, so $D_1^\top = -D_1$ and $\mathbf{a}^\top D_1 \mathbf{b} = -\mathbf{b}^\top D_1 \mathbf{a}$ for any $\mathbf{a}, \mathbf{b}$. Applying this with $\mathbf{a} = \mathbf{u}$, $\mathbf{b} = \mathbf{u} \odot \mathbf{u}$ gives $\mathbf{u}^\top D_1(\mathbf{u} \odot \mathbf{u}) = -(\mathbf{u} \odot \mathbf{u})^\top D_1 \mathbf{u} = -\sum_i u_i^2 (D_1 \mathbf{u})_i$, while the first bracket contracted with $\mathbf{u}$ is term by term $\sum_i u_i^2 (D_1 \mathbf{u})_i$. The two cancel. $\square$

We verified this to machine precision ($\sim 10^{-17}$) with random vectors at every size used here, and separately verified that Eq. (7) agrees with $-u \partial_x u$ to the expected $\mathcal{O}(\Delta x^2)$ truncation order. The naive form fails the exact-zero test for generic $\mathbf{u}$ by an $\mathcal{O}(1)$ amount that grows with n_x : in a KdV run it produced an 11% drift in $\|\mathbf{u}(t)\|_2$ over the simulated interval, for an equation that conserves it exactly.

With the closure of Eq. (7), $\frac{d}{dt}\|\mathbf{u}\|_2^2 = 2\mathbf{u}^\top F_1 \mathbf{u}$ for both PDEs at any resolution. For Burgers this is ≤ 0 , so $\|\mathbf{u}(t)\|_2$ is non-increasing; for KdV, $F_1 = -\delta D_3$ is itself antisymmetric by the same argument, so it vanishes and $\|\mathbf{u}(t)\|_2$ is exactly conserved.

C. Exact conservation of the ODE does not survive time discretization

Proposition 1 concerns the exact flow of Eq. (6) and says nothing about a time-stepping scheme. Take one forward-Euler step $\mathbf{u}_{n+1} = \mathbf{u}_n + \Delta t f(\mathbf{u}_n)$ with $f(\mathbf{u}) = F_1 \mathbf{u} + F_2(\mathbf{u} \otimes \mathbf{u})$ the KdV right-hand side, for which $\mathbf{u}^\top f(\mathbf{u}) = 0$ exactly. Then

$$\begin{aligned} \|\mathbf{u}_{n+1}\|_2^2 &= \|\mathbf{u}_n\|_2^2 + 2\Delta t \mathbf{u}_n^\top f(\mathbf{u}_n) + \Delta t^2 \|f(\mathbf{u}_n)\|_2^2 \\ &= \|\mathbf{u}_n\|_2^2 + \Delta t^2 \|f(\mathbf{u}_n)\|_2^2 \geq \|\mathbf{u}_n\|_2^2. \end{aligned} \quad (8)$$

Forward Euler therefore *provably* injects energy into KdV's exactly conservative ODE on every step, however small Δt is. We confirmed Eq. (8) numerically to 10^{-16} relative precision. The remedy, with its own exactness proof, is the implicit midpoint rule (Sec. VD); the observed consequence of ignoring it is a monotone growth $\|\mathbf{u}\|_2 = 1.00 \rightarrow 1.19 \rightarrow 1.42 \rightarrow 1.72$ over three steps.

D. The Reynolds number sets the resolution both digital routes need

Non-dimensionalizing Eq. (1) by amplitude $U = \|u_0\|_\infty$ and domain length $L = 1$ leaves the single parameter $\text{Re} = UL/\nu$. Proposition 1 guarantees an energy bound but not monotonicity; central differences on the convective term are monotonicity-preserving only if the cell Reynolds number satisfies $\text{Re}_\Delta = U\Delta x/\nu \lesssim 2$. We confirmed this bound two independent ways: by linearizing the discrete update near $u \approx U$ and requiring the off-diagonal coefficients to stay non-negative, and by the spectral analysis of Sec. IV. Table II evaluates both numbers for the systems used here.

III. ROUTE 1: CARLEMAN-QSVT, AS BUILT

This section reports what we constructed and measured. Its asymptotic cost is deferred to Sec. IV so that measured and projected quantities are never mixed.

TABLE II. Reynolds and cell Reynolds numbers for the Burgers systems solved in this work ($L = 1$). The variational grid sits a factor of six above the monotonicity threshold, a coarse-grid overshoot we report rather than conceal; $\text{Re}_\Delta \leq 2$ at $\nu = 0.01$ requires $n_x \gtrsim 50$.

System	ν	Re	Re_Δ
Route 1, Carleman ($n_x = 5$)	0.1	9.5	1.9
Route 2, variational ($n_x = 8$)	0.01	100	12.5
Threshold $\text{Re}_\Delta \leq 2$ requires	0.01	100	$n_x \gtrsim 50$

A. The Carleman lift

Carleman linearization introduces the tensor hierarchy $\mathbf{y} = (\mathbf{u}, \mathbf{u}^{\otimes 2}, \dots, \mathbf{u}^{\otimes N_C})^\top$, on which Eq. (6) becomes the *linear* system

$$\dot{\mathbf{y}} = \mathbf{A}\mathbf{y} + \mathbf{b}, \quad (9)$$

with $\mathbf{A}$ block upper-bidiagonal: the diagonal block A_{jj} propagates $\mathbf{u}^{\otimes j}$ under F_1 and the superdiagonal block $A_{j,j+1}$ couples it to $\mathbf{u}^{\otimes(j+1)}$ through F_2 . Truncating at order N_C (setting $\mathbf{u}^{\otimes(N_C+1)} \equiv 0$) closes the system at dimension $\sum_{j=1}^{N_C} n_x^j$.

The convergence theorem for this truncation [3] requires F_1 to be strictly dissipative and controls the error through the ratio $R = \|\mathbf{u}_0\|_2 \|F_2\|_2 / |\text{Re } \mu_1(F_1)|$, with μ_1 the eigenvalue of largest real part. For our *periodic* construction this theorem does not literally apply: the periodic Laplacian has a zero (DC) mode, so $\text{Re } \mu_1(F_1) = 0$ exactly and R diverges. The same is true a fortiori for KdV, where F_1 is skew-symmetric. We therefore validated the truncation order empirically instead, and report it as such: $N_C = 2$ and $N_C = 3$ give near-identical RMSE against a high-accuracy RK45 reference, so at these parameters the truncation is not what limits accuracy; the dispersion stiffness $\delta/\Delta x^3$ at fixed Δt is, which is why δ is kept small. Section IV B returns to what the convergence condition costs when it *is* enforced.

B. Time marching and block encoding

Each timestep is a linear solve. We use Crank–Nicolson (trapezoidal) marching,

$$\left(I - \frac{\Delta t}{2} \mathbf{A}\right) \mathbf{y}_{n+1} = \left(I + \frac{\Delta t}{2} \mathbf{A}\right) \mathbf{y}_n, \quad (10)$$

rather than implicit Euler: the block-encoded matrix is the same, but the local error is $\mathcal{O}(\Delta t^2)$ instead of $\mathcal{O}(\Delta t)$, which we checked classically to need roughly $20\times$ fewer timesteps at matched accuracy, following the higher-order rescaling analysis of Ref. [13].

We implemented and compared two block encodings of the Crank–Nicolson matrix: a dense arbitrary-unitary construction, and one built from the matrix's Pauli decomposition as a linear combination of unitaries (LCU).

TABLE III. Route 1 systems solved ($N_C = 2$, $\Delta t = 0.1$, $t \in [0, 0.3]$). κ_Q is the measured effective condition number of the encoded matrix, which sets the required QSVT degree.

PDE	n_x	$\dim A$	κ_Q (dense)	κ_Q (LCU)
Burgers ($\nu = 0.1$)	5	30	9.35	8.79
KdV ($\delta = 0.0625$)	4	20	4.02	4.75

The comparison produced a result worth stating explicitly, because it inverts the usual intuition. The Pauli-LCU encoding is the *more* expensive of the two for this matrix family, by two to three orders of magnitude, because the Kronecker-shift structure of Eq. (9) is sparse in the computational basis but *dense* in the Pauli basis. Sparsity in one basis buys nothing if the encoding reads the other.

C. QSVT inversion, and what we verified

Matrix inversion is performed by QSVT [7] applied to the block encoding, with phase angles generated for a target condition number $\kappa = 8$ (561 angles). Table III lists the systems solved and Table IV their exact transpiled cost.

Every quantum result agrees with its classical linear-algebra counterpart to numerical precision. Both notebooks compare three solvers — the exact untruncated nonlinear ODE integrated by RK45, the Carleman-truncated system solved exactly (isolating truncation error), and the Carleman-truncated system solved by QSVT under both encodings — so a disagreement would localize to a specific stage. None appeared.

The gate counts in Table IV should be read as a property of the *encoding we chose*, not of Carleman-QSVT as such, and the distinction is worth stating plainly. Our dense construction assembles the full $2^{n_w} \times 2^{n_w}$ dilation $\begin{bmatrix} A_s & \sqrt{I - A_s A_s^\dagger} \\ \sqrt{I - A_s^\dagger A_s} & -A_s^\dagger \end{bmatrix}$ explicitly and hands it to the transpiler as an opaque unitary. By that point the Carleman matrix’s sparsity and Kronecker-shift structure have been destroyed: the transpiler sees an arbitrary unitary and falls back on generic synthesis [16], which is precisely the $\mathcal{O}(4^{n_w})$ cost we measure in Sec. IV C. The QSVT projector layers, by contrast, are built as multi-controlled phase gates rather than dense diagonal matrices, so the blow-up is localized to the block-encoding layers alone. What a structure-aware encoding would cost instead is the subject of Sec. IV C, and is the reason the subnormalization analysis there matters. Figure 2 shows the resulting Burgers propagation, Fig. 3 the per-timestep RMSE, post-selection success probability, and amplitude-encoding fidelity, and Fig. 4 the classical space-time reference.

TABLE IV. **Measured.** Exact transpiled cost of one QSVT matrix inversion, $\kappa = 8$ (561 phase angles), $\{cx, u_3\}$ basis. This is paid at *every* timestep.

PDE	encoding	qubits	gates	CX gates
Burgers	dense	6	2,612,767	996,697
Burgers	Pauli-LCU	14	1,716,882,557	803,286,678
KdV	dense	6	2,610,531	996,697
KdV	Pauli-LCU	13	396,632,217	185,505,488

IV. WHAT ROUTE 1 COSTS

We now derive the asymptotic cost of the pathway. Every quantity in this section is *projected*, not measured; where possible the projections are calibrated against published gate-level data for a 30×30 Burgers instance [1], and we state the agreement. Our reconstructed data-element count, subnormalization, effective condition number, and success probability all reproduce the published values — the success-probability model to 5% and the subnormalization formula exactly — which is what licenses extrapolating them.

A. The error budget, and which errors are amplified

The pipeline accumulates five error sources: spatial discretization, Carleman truncation, time discretization, QSVT polynomial approximation, and statistical (shot) error. They are not interchangeable. Errors incurred *before* the linear solve are amplified by the effective condition number κ_Q when propagated through it, while errors incurred *by* the solve are not. This distinction determines the budget allocation and is why a seemingly modest input error can dominate the final cost.

B. The binding constraint: Carleman convergence requires $\text{Re} < \pi/2$

This is the deepest limitation of the route, and it is a statement about the physics, not the hardware. The standard analysis [3, 4] of the quadratic system requires F_1 dissipative and controls the truncation error by

$$R := \frac{\|\mathbf{u}_0\|_2 \|F_2\|_2}{|\text{Re } \mu_1(F_1)|}, \quad N_C = \left\lceil \frac{\ln(1/\epsilon)}{\ln(1/R)} \right\rceil, \quad (11)$$

valid only for $R < 1$; if $R \geq 1$ no such bound exists and increasing N_C is not guaranteed to help.

Evaluated naively for the Dirichlet discretization, $R \simeq \text{Re } n_x^{3/2} / 2\pi^2$ grows with grid refinement — which cannot be physical, since refining a grid does not make a PDE more nonlinear. The inflation comes from using the unweighted operator norm of F_2 , which measures its action

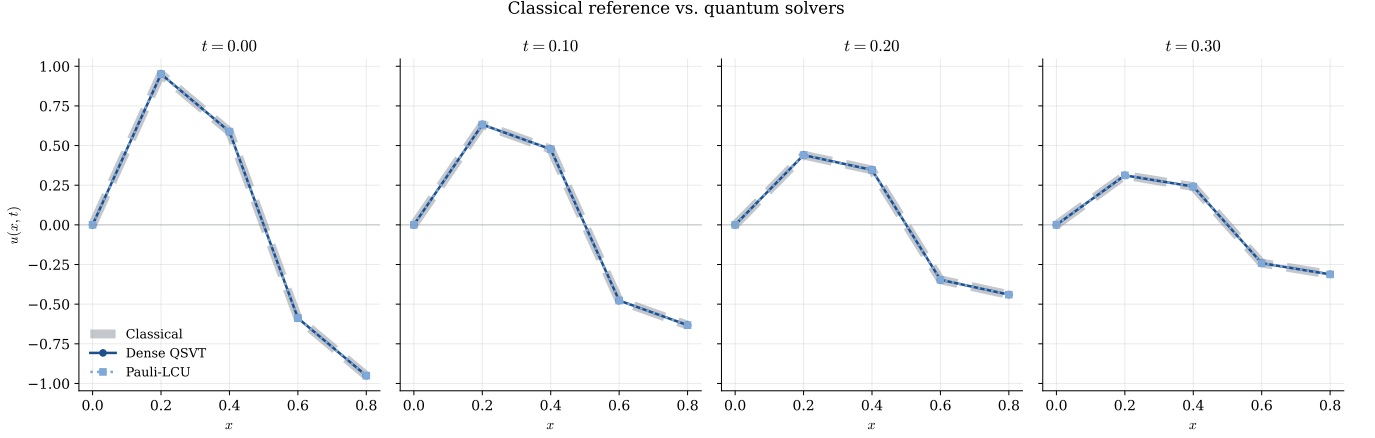


FIG. 2. Burgers propagation via Route 1: classical (dashed) versus dense QSVT (solid) versus Pauli-LCU QSVT (dotted). At $\nu = 0.1$ diffusion dominates advection, so the profile decays and mildly skews rather than steepening into a shock. The three curves coincide to numerical precision, which is the point of the figure: both block encodings implement the same Carleman-lifted operator A correctly, not merely a plausible one.

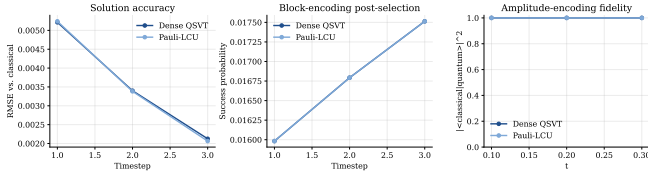


FIG. 3. Per-timestep RMSE against the classical reference, QSVT post-selection success probability, and amplitude-encoding fidelity, for both block encodings. Success probability is the quantity that compounds over a sequential run, and Sec. IV D shows why that compounding is fatal without a dilation.

on all of $\mathbb{R}^{n_x^2}$, whereas the dynamics only ever visits the tensor manifold $\{\mathbf{u} \otimes \mathbf{u}\}$. Restricting to that manifold, $[F_2 \mathbf{u}^{\otimes 2}]_j \approx -u_j (\partial_x u)_j$, so $\|F_2 \mathbf{u}^{\otimes 2}\|/\|\mathbf{u}\| \leq \|\partial_x u\|_\infty$, which is grid-independent. With $\mu_1(F_1) \rightarrow -\pi^2 \nu / L^2$ as $\Delta x \rightarrow 0$ and $\|\partial_x u\|_\infty = 2\pi U / L$ for $u_0 = U \sin(2\pi x / L)$, the sharpened ratio is

$$R_\star = \frac{\|\partial_x u\|_\infty}{\pi^2 \nu / L^2} = \frac{2}{\pi} \text{Re},$$

$$\implies R_\star < 1 \iff \text{Re} < \frac{\pi}{2} \approx 1.57. \quad (12)$$

The conclusion is robust to how it is computed: for a published $\text{Re} = 100$ instance the naive estimate gives $R = 56.6$, the sharpened estimate $R_\star = 63.7$, and direct evaluation of Eq. (11) from the assembled matrices gives $R = 76.2$. All three place that instance two orders of magnitude outside the provably convergent regime. Figure 5(a) shows N_C diverging as $\text{Re} \rightarrow \pi/2$.

Remark 1 (Rescaling does not help). Rescaling $\mathbf{u} \rightarrow \mathbf{u}/\Gamma$ to shrink $\|\mathbf{u}_0\|$ leaves Eq. (6) satisfied with $F_2 \rightarrow \Gamma F_2$, so the product $\|\mathbf{u}_0\| \|F_2\|$ in Eq. (11) is unchanged. The constraint is physical, not a normalization artifact.

Burgers: classical reference, space-time surface

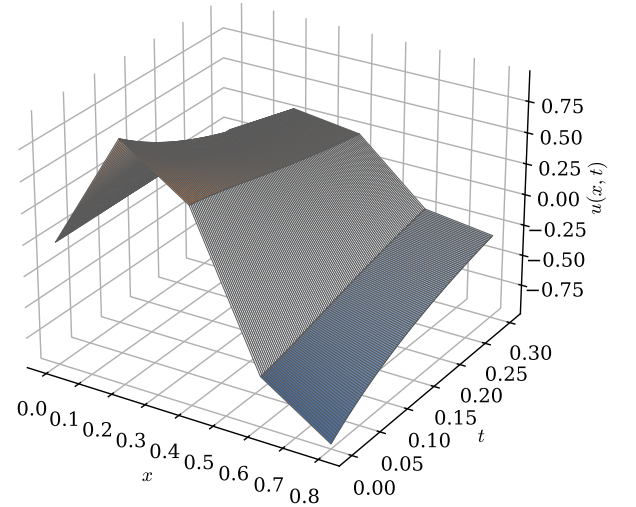


FIG. 4. Classical space-time reference solution for the Burgers instance of Table III (finely sub-stepped, for visualization). Both digital routes are checked against this surface at every timestep.

Remark 2 (KdV is worse, not better). For KdV, F_1 is skew-symmetric and $\text{Re} \mu_1 = 0$ exactly, so $R = \infty$ and Eq. (11) is vacuous. One must fall back on finite-horizon bounds of the form $\varepsilon_{\text{CL}} \lesssim \|\mathbf{u}_0\| (T/\tau_{\text{nl}})^{N_C}$ with $\tau_{\text{nl}} = \|\partial_x u\|_\infty^{-1}$ [5], which converge only for T less than roughly one eddy turnover. Restarting is not an escape, because re-preparing an amplitude encoding of $\mathbf{u}(T)$ requires tomography.

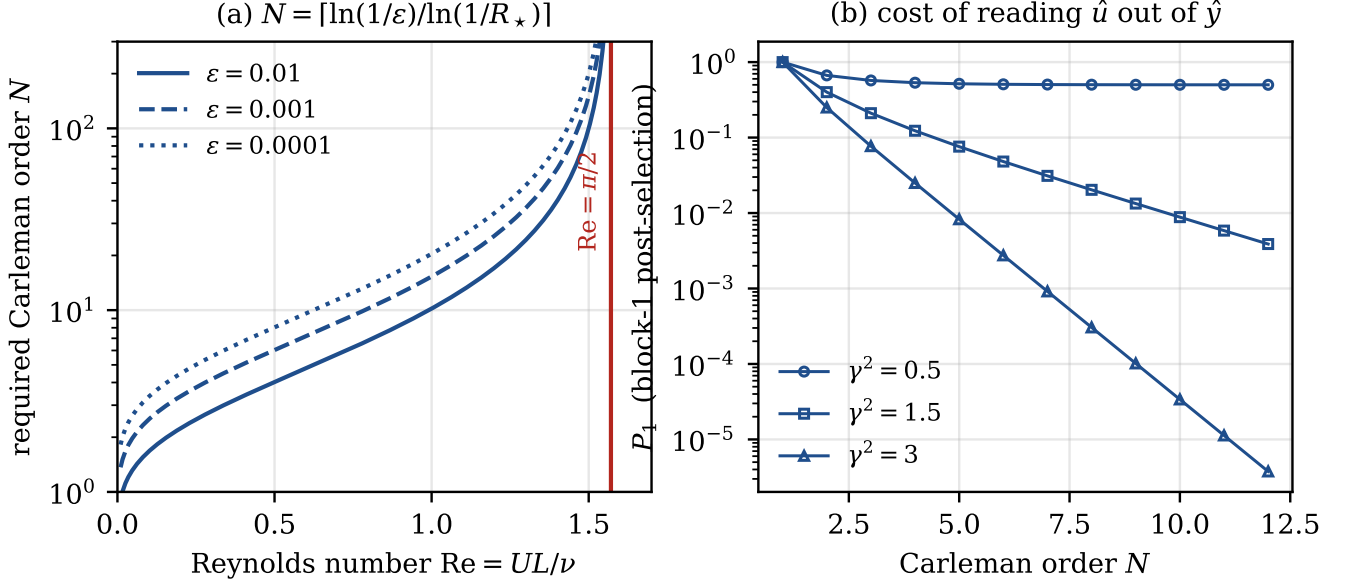


FIG. 5. **Projected.** (a) Carleman order N_C required by Eq. (11) to reach relative error ϵ , using the sharpened ratio $R_\star = 2Re/\pi$. Beyond $Re = \pi/2$ (red line) the series has no proven convergence regime and no finite N_C suffices — this is the binding constraint on the entire route. (b) The block-1 post-selection probability, which decays geometrically in N_C unless the state is rescaled so that $\gamma = \|\mathbf{u}\|_2 < 1$.

C. The block encoding, not the matrix, is the largest avoidable cost

For the coherent-permutation block encoding, the subnormalization obeys

$$\alpha_{cp} \simeq n_x^{N_C-1} \lambda_{cfl}, \quad \lambda_{cfl} = \Delta t / \Delta x, \quad (13)$$

which we confirmed numerically [Fig. 6]. Because α enters the cost linearly through $\kappa_Q = \alpha/\sigma_{\min}$, the algorithm is polynomial in grid size and *exponential in the Carleman order* — precisely the scaling the Carleman embedding was supposed to avoid. Replacing it with an arithmetic sparse-access oracle, which indexes the $\mathcal{O}(N_C)$ nonzero entries per row directly rather than synthesizing a dense unitary, restores

$$\alpha_{sp} = \mathcal{O}(N_C^2 \lambda_{diff}), \quad (14)$$

independent of n_x except through the diffusion number. The gap is already a factor of 17 at $n_x = 15$, $N_C = 3$. This is an engineering problem with a known fix, and we treat it as such below.

A complementary and independently measured statement holds for our own dense construction of Sec. III: its per-layer gate count fits $\propto 2^{2.03n_w} = \mathcal{O}(4^{n_w})$ against its own qubit count n_w , the generic cost of synthesizing an arbitrary unitary with no exploitable structure [16], giving $\mathcal{O}(n_x^4)$ in grid size at $N_C = 2$. The two statements identify the same lesson from opposite directions, one asymptotic and one measured: the encoding, not the matrix, dominates. A structure-blind encoding pays $\mathcal{O}(4^{n_w})$

in synthesis; a structure-aware but subnormalization-blind one pays $n_x^{N_C-1}$ in QSVT degree. Neither cost is intrinsic to the Carleman lift, which is why we regard both as engineering problems rather than obstructions — in sharp contrast to the convergence threshold of Sec. IV B, which is neither.

D. Sequential time stepping does not scale, and the fix has a bonus

Solving one implicit step at a time — the natural reading of Eq. (10), and what our own implementation does — is not viable asymptotically. Each solve is heralded with probability $p_s < 1$, and the output of one step is the input of the next, so chaining m steps costs $\mathcal{O}(p_s^{-m/2})$ even with optimal amplitude amplification. This is the practical meaning of the success-probability panel in Fig. 3.

The remedy is the dilated “history-state” formulation, which solves for all timesteps at once in a single enlarged linear system. It comes with a genuinely useful surprise: the product αm entering the dilated system’s condition number is *independent of Δt* . Refining the time step multiplies the number of steps and divides the subnormalization by the same factor, so nothing is lost. This cancellation is what makes the assembled estimate below finite at all.

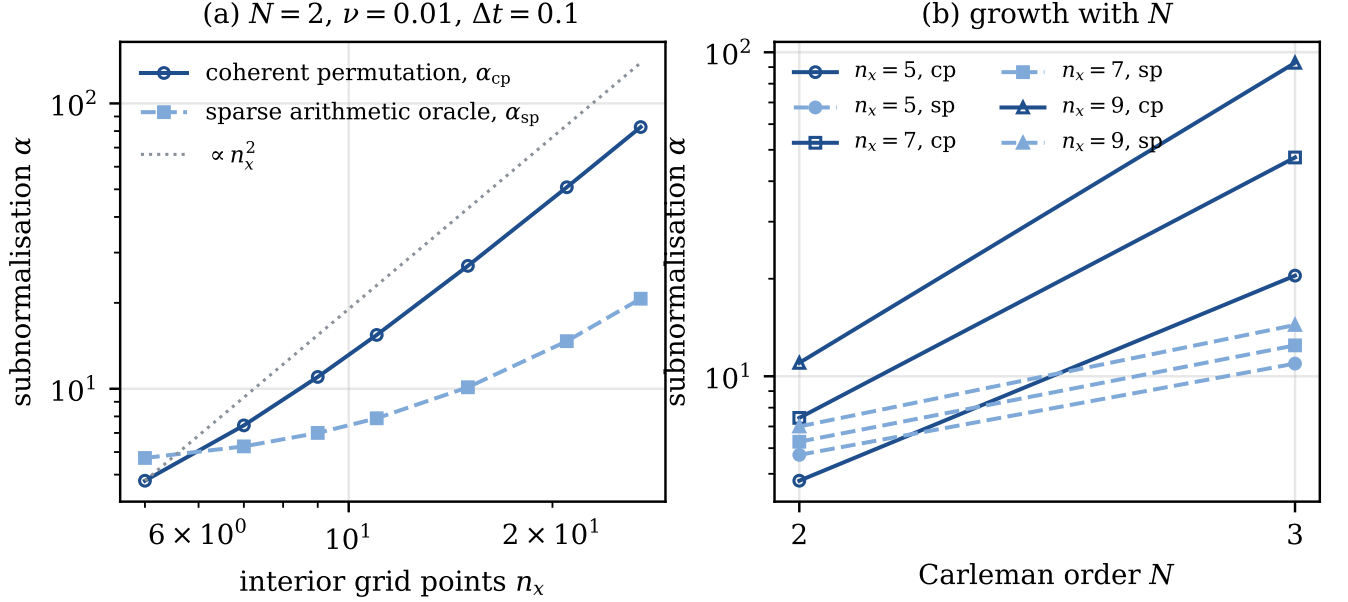


FIG. 6. Growth of the block-encoding subnormalization α , which sets the effective condition number and hence the QSVT degree. (a) Versus grid size at $N_C = 2$: the coherent-permutation encoding (cp) tracks $\propto n_x^2$ while the sparse arithmetic oracle (sp) grows far more slowly. (b) Versus Carleman order: cp grows geometrically in N_C , sp does not. Equations (13)–(14) are the corresponding closed forms. Neither encoding is constructed as a circuit here: α is evaluated exactly from the assembled Carleman matrix (α_{cp} as the sum of $|\cdot|$ over distinct value/diagonal-offset pairs, $\alpha_{sp} = s\|L\|_{\max}$), so these are exact cost-model constants rather than measured circuit resources. The circuits we did build are the dense and Pauli-LCU encodings of Table IV.

E. Assembled scaling

Combining the above under stated assumptions — sparse-oracle block encoding, high-order time integration, history-state dilation, normalized initial data, $\text{Re} < \pi/2$ so that N_C is finite, and a grid that resolves the solution — the sequential two-qubit depth is

$$D_{\text{QSVT}} = \tilde{O}\left(\frac{\kappa_V T N_C^3}{\text{Re } \epsilon}\right), \quad (15)$$

with

$$N_C = \Theta\left(\frac{\log(1/\epsilon)}{\log(\pi/2\text{Re})}\right), \quad \#\text{qubits} = \Theta\left(\frac{\log^2(1/\epsilon)}{\log(\pi/2\text{Re})}\right). \quad (16)$$

The depth model underlying Eq. (15) is calibrated against published gate-level depths for four independent instances in Fig. 7, and reproduces them within a factor of two. Table V collects the full breakdown.

F. A worked instance, against a laptop

Table VI evaluates Eq. (15) at the largest Reynolds number for which the Carleman series provably converges, and places it beside a classical pseudo-spectral solve of the same instance.

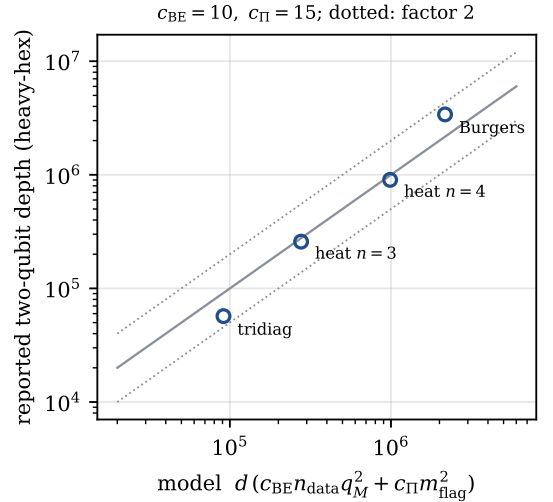


FIG. 7. Calibration of the depth model against published two-qubit depths for four independent circuit instances [1]. Dotted lines mark a factor of two. The agreement is what licenses extrapolating the model to the regimes of Table VI.

The gap is roughly 10^6 in operation count before error correction and 10^9 – 10^{10} after. Both sides scale as $\tilde{O}(\epsilon^{-1})$ in accuracy; the quantum side carries additional factors $\kappa_V N_C^3$, its depth is inherently sequential where classical FFTs parallelize trivially, and it returns a quantum state

TABLE V. **Projected.** Assembled resource scaling for Route 1 under the assumptions of Sec. IV E. κ_V , the eigenvector condition number of the Carleman matrix, is the least controlled quantity here: A is strongly non-normal, and we are not aware of a bound on κ_V in the literature.

Resource	Scaling
Logical qubits	$\Theta(N_C \log n_x + \log m)$
Carleman order N_C	$\Theta(\log(1/\epsilon)/\log \frac{\pi}{2\text{Re}})$
Grid points n_x	$\Theta(\max(\text{Re}, \epsilon^{-1/2}))$
Effective κ_Q	$\mathcal{O}(\kappa_V N_C T / (\text{Re} \epsilon))$
Polynomial degree d	$\tilde{\mathcal{O}}(\kappa_Q)$
Total depth	$\tilde{\mathcal{O}}(\kappa_V T N_C^3 \text{Re}^{-1} \epsilon^{-1})$
Shots, observable	$\mathcal{O}(\epsilon_{\text{obs}}^{-1})$
Shots, full field	$\mathcal{O}(n_x \epsilon^{-2})$

TABLE VI. **Projected.** A worked instance at $\text{Re} = 1$ (so $R_* = 0.637$), $\epsilon = 10^{-3}$, $T = 0.3$, $n_x = 128$. Depths are logical two-qubit gates; the required gate infidelity implies full fault tolerance, with a further 10^3 – 10^4 physical-qubit overhead per logical qubit.

Quantity	Value
Carleman order N_C	16
System dimension	2^{112}
Timesteps m	4915
Subnormalization α_{sp}	2.0×10^3
Dilated condition number	1.0×10^7
Polynomial degree d	2.3×10^8
Total depth	1.3×10^{12}
Total logical qubits	~ 135
Required gate infidelity	10^{-12}
Runtime at 100 ns/gate	~ 36 hours
<i>Classical pseudo-spectral, same instance</i>	
Arithmetic operations	4.4×10^6
Memory	1 kB
Runtime	~ 1 ms

rather than a field.

One structural quantum win survives intact: the qubit count $\mathcal{O}(N_C d_{\text{sp}} \log n_x)$ in d_{sp} spatial dimensions is genuinely exponentially smaller than the classical $\mathcal{O}(n_x^{d_{\text{sp}}})$, and nothing in the analysis erodes it. Comparing work rather than memory, a crossover requires $d_{\text{sp}} \geq 3$ before constants, and is conditional on all of $\text{Re} = \mathcal{O}(1)$, observable-only readout, and κ_V polynomially bounded.

G. Why Burgers alone cannot settle the question

There is an uncomfortable point specific to Burgers. The Cole–Hopf transformation $u = -2\nu \partial_x \ln \phi$ maps Eq. (1) *exactly* onto the linear heat equation $\partial_t \phi =$

$\nu \partial_{xx} \phi$. Burgers is therefore not a genuinely nonlinear test problem at all: it is a linear problem in disguise, solvable classically in $\mathcal{O}(n_x \log n_x)$ total — one FFT, a diagonal multiplication, one inverse FFT — for *any* T and *any* Re , with no time stepping whatsoever.

This does not invalidate Burgers as a benchmark for checking that circuits compile and produce the right answer, which is how we use it. But it does mean a Burgers demonstration cannot by itself be evidence that a method handles nonlinearity. The correct control is a PDE with no linearizing substitution, which is exactly why we carry KdV through every stage of this work. We note in passing that Cole–Hopf does not straightforwardly rescue the quantum route either: recovering $|u\rangle$ from $|\phi\rangle$ requires the nonlinear map $\phi \mapsto \partial_x \ln \phi$ on amplitudes, which is not implementable without tomography.

V. ROUTE 2: VARIATIONAL AMPLITUDE-ENCODED PROPAGATION

Route 1 pays for linearity in circuit depth. Route 2 declines to pay: the nonlinearity is evaluated by a classical optimizer, and the quantum register never holds anything larger than the field itself. Everything in this section is measured.

A. Encoding, ansatz, and cost function

The field is represented as $\mathbf{u} = \text{amp} \cdot \boldsymbol{\psi}(\boldsymbol{\theta})$, with $\boldsymbol{\psi}(\boldsymbol{\theta})$ the real, unit-norm statevector of an n -qubit **RealAmplitudes** ansatz (n repetitions of R_y rotations with linear CNOT entanglement) and $\text{amp} \in \mathbb{R}$ tracked classically. The ansatz has exactly $P = n(n+1)$ parameters. Each timestep minimizes

$$C(\boldsymbol{\theta}, \text{amp}) = \|\text{amp} \cdot \boldsymbol{\psi}(\boldsymbol{\theta}) - \mathbf{u}_{\text{target}}\|_2^2, \quad (17)$$

$$\mathbf{u}_{\text{target}} = \mathbf{u}_{\text{old}} + \Delta t f(\mathbf{u}_{\text{rhs}}), \quad (18)$$

where $\mathbf{u}_{\text{rhs}} = \mathbf{u}_{\text{old}}$ for forward Euler (Burgers) or the implicit midpoint state (KdV, Sec. V D). Writing $C_0(\boldsymbol{\theta}) = \langle \boldsymbol{\psi}(\boldsymbol{\theta}) | \mathbf{u}_{\text{target}} \rangle$ and minimizing over amp at fixed $\boldsymbol{\theta}$ gives $\text{amp}^* = C_0$; the parameters then follow *natural* gradient descent,

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta M(\boldsymbol{\theta})^+ \nabla_{\boldsymbol{\theta}} [-2 \text{amp}^* C_0(\boldsymbol{\theta})], \quad (19)$$

with M^+ the pseudo-inverse of the Fubini–Study metric and η set by backtracking line search. This is the standard cost-function form of the McLachlan principle [8, 10]. What is specific to this work is that every ingredient of Eq. (19) is measured by a shallow circuit with at most one ancilla and no controlled-ansatz operations anywhere.

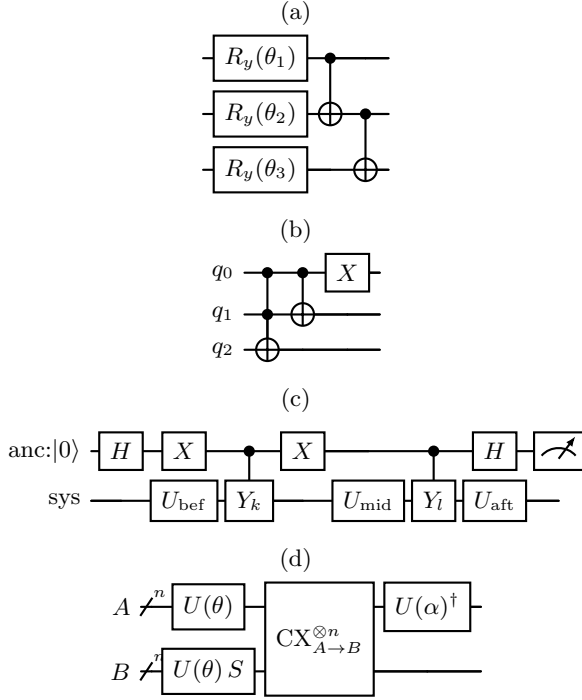


FIG. 8. Circuit primitives of Route 2: (a) one repetition of the R_y/CNOT ansatz layer, applied n times in total; (b) the ancilla-free cyclic shift S (shown for $n = 3$; the staircase of $(n-1-k)$ -controlled X gates extends to any n); (c) the single-ancilla metric off-diagonal circuit; (d) the two-register nonlinear cross-term circuit, with no controlled-ansatz gates anywhere. Exact transpiled costs are in Table VIII.

B. Circuit-level measurement

Figure 8 shows the four primitives. For a single-Pauli-rotation parameter the metric diagonal is $M_{kk} = 1/4$ exactly, needing no circuit; off-diagonal entries come from a single-ancilla circuit inserting the two generators at the relevant gate positions and reading $\langle X \rangle$, which we verified against a finite-difference evaluation of the exact Fubini–Study metric to $\sim 10^{-10}$. The overlaps entering $\mathbf{u}_{\text{target}}$ — identity, each shift term of F_1 , and the building blocks $\chi_i^\pm = (S^{\pm 1}\psi)_i\psi_i$ and $\text{sq}_i^\pm = ((S^{\pm 1}\psi)_i)^2$ required by the skew closure — are each measured by a compute–uncompute circuit whose all-zero outcome probability equals the squared overlap, $P(0) = g^2$. The cyclic shift S is implemented ancilla-free by a ripple-carry Toffoli chain.

C. The sign problem, and a certificate that solves it

Because $P(0) = g^2$, these circuits return $|g|$ and not its sign. Two consequences follow, and both required more than a heuristic fix.

First, the parameter-shift derivative $\partial g / \partial \theta_l = [P(0)(\theta_l + \frac{\pi}{2}) - P(0)(\theta_l - \frac{\pi}{2})] / 4g$ is genuinely singular at $g = 0$: $P(0)$ has a double root there, so the numerator is always $\mathcal{O}(g^2) \rightarrow 0$ regardless of the true derivative, and

naively reading this as “derivative = 0” silently discards real gradient information. We measured this occurring on $\sim 8\%$ of evaluations in a real optimization run — not a rare edge case. Below a small threshold on $|g|$ we substitute a classical finite difference of the exact overlap.

Second, the sign itself must be tracked across the optimization. Two heuristics we tried — recomputing only when the current magnitude looked small, and freezing the sign across a line search’s candidate points — both silently corrupted the KdV optimization (9–19% relative error, up from $\sim 10^{-8}$), because a term can cross zero between two evaluations without ever being observed near zero at a check point. The following certificate removes the guesswork.

Proposition 2 (exact single-parameter form). *For fixed reference states and a single rotation angle θ_l , all other parameters held fixed, $g(\theta_l) = A \cos(\theta_l/2) + B \sin(\theta_l/2)$ exactly, for constants A, B independent of θ_l .*

Proof. $R_y(\theta_l) = \cos(\theta_l/2)I + \sin(\theta_l/2)K$ for a fixed real generator K . Writing the rest of the ansatz as fixed unitaries, $\psi(\theta_l) = \cos(\theta_l/2)\mathbf{a} + \sin(\theta_l/2)\mathbf{b}$ for fixed $\mathbf{a}, \mathbf{b}$; contracting with a fixed reference $\mathbf{v}$ gives the claimed form with $A = \langle \mathbf{v} | \mathbf{a} \rangle$, $B = \langle \mathbf{v} | \mathbf{b} \rangle$. $\square$

Since $A^2 + B^2 \leq \|\mathbf{v}\|^2$ and every reference vector used here has $\|\mathbf{v}\|_2 \leq 1$, Proposition 2 yields the uniform bound $|\partial g / \partial \theta_l| \leq 1/2$, verified numerically to 10^{-10} . Hence:

Proposition 3. *If $|g(\theta_{\text{last}})| > \frac{1}{2}\|\theta_{\text{new}} - \theta_{\text{last}}\|_1$, then g cannot vanish anywhere on the straight line between θ_{last} and θ_{new} .*

Proof. Along $\theta(t) = \theta_{\text{last}} + t(\theta_{\text{new}} - \theta_{\text{last}})$, the chain rule and the uniform bound give $|\frac{d}{dt}g| \leq \frac{1}{2}\|\theta_{\text{new}} - \theta_{\text{last}}\|_1$. Integrating, $|g(\theta(t)) - g(\theta_{\text{last}})| \leq t \cdot \frac{1}{2}\|\theta_{\text{new}} - \theta_{\text{last}}\|_1$, which at $t = 1$ is smaller than $|g(\theta_{\text{last}})|$ by hypothesis, so g cannot reach zero on the segment. $\square$

This is a proven certificate rather than a heuristic: a sign is carried forward at zero circuit cost whenever it holds, and falls back to one classical evaluation only when it fails. On a Burgers run this reduced classical calls from one full $\mathcal{O}(2^n)$ lookup per trial point — thousands per timestep — to ~ 55 per timestep, matching the always-classical baseline to full numerical precision on both PDEs and multiple random seeds.

D. Implicit midpoint for KdV

Section II C proved forward Euler injects energy into KdV. Implicit midpoint, $\mathbf{u}_{n+1} = \mathbf{u}_n + \Delta t f(\frac{\mathbf{u}_n + \mathbf{u}_{n+1}}{2})$, does not.

Proposition 4. *Implicit midpoint conserves $\|\mathbf{u}\|_2^2$ exactly, for any Δt , whenever $\mathbf{v}^\top f(\mathbf{v}) = 0$ for all $\mathbf{v}$.*

TABLE VII. **Measured.** Route 2 systems solved ($\Delta t = 0.1$, 150 natural-gradient iterations per step).

PDE	n	P	rel. L^2 error, ideal / noisy
Burgers ($\nu = 0.01$)	3	12	$2-4 \times 10^{-6}$ / 0.17–0.20
KdV ($\delta = 0.0625$)	2	6	2×10^{-8} – 2×10^{-7} / 0.07–0.13

TABLE VIII. **Measured.** Exact transpiled cost of each primitive of Fig. 8, one instance, $\{cx, u_3\}$ basis, at the qubit count actually used.

PDE	circuit	qubits	gates	depth
Burgers ($n = 3$)	identity/linear	3	128	68
Burgers ($n = 3$)	nonlinear	6	351	148
Burgers ($n = 3$)	metric	4	85	40
KdV ($n = 2$)	identity/linear	2	29	17
KdV ($n = 2$)	nonlinear	4	140	70
KdV ($n = 2$)	metric	3	55	33

Proof. With $\mathbf{u}_{\text{mid}} = \frac{1}{2}(\mathbf{u}_n + \mathbf{u}_{n+1})$ we have $\mathbf{u}_{n+1} - \mathbf{u}_n = \Delta t f(\mathbf{u}_{\text{mid}})$, so $\|\mathbf{u}_{n+1}\|_2^2 - \|\mathbf{u}_n\|_2^2 = (\mathbf{u}_{n+1} - \mathbf{u}_n)^\top (\mathbf{u}_{n+1} + \mathbf{u}_n) = 2\Delta t \mathbf{u}_{\text{mid}}^\top f(\mathbf{u}_{\text{mid}}) = 0$. $\square$

We solve the small classical n_x -dimensional nonlinear equation for $\mathbf{u}_{n+1}$ once per timestep, fit a second ansatz encoding to $\mathbf{u}_{\text{mid}}$, and use it as $\mathbf{u}_{\text{rhs}}$ in Eq. (18) while $\mathbf{u}_{\text{old}}$ still supplies the identity term. This holds $\|\mathbf{u}(t)\|_2 = 1.0000000000$ at every step classically, against $1.00 \rightarrow 1.19 \rightarrow 1.42 \rightarrow 1.72$ under forward Euler, and to $\sim 10^{-7}$ for the quantum-fitted state.

E. Results and measured resources

Tables VII and VIII summarize both systems and the exact transpiled cost of each primitive. Figures 9 and 10 show the propagation.

We separate measured regimes explicitly, as PRX Quantum expects. The *ideal* curves are noiseless statevector simulation. The *noisy* curves use the **FakeSherbrooke** calibration snapshot of an IBM superconducting processor, with device noise and 4000-shot readout — no error mitigation, no post-selection. Because only $\sqrt{P(0)}$ is measurable this way, the noisy runs take each overlap’s sign from the ideal simulation; this is a stated limitation of the hardware-noise demonstration, not of the method, since Proposition 3 supplies the sign at zero circuit cost whenever its condition holds.

F. Measured scaling

We measured how Route 2’s resources grow by building and transpiling the actual circuits at $n = 2, \dots, 6$ ($n_x = 4, \dots, 64$) against the **FakeSherbrooke** calibration

snapshot at optimization level 3, rather than evaluating a closed-form asymptotic alone. Every quantity is polynomial in n and therefore *polylogarithmic* in grid size: the parameter count is exactly $P = n(n+1) = \Theta(\log^2 n_x)$ at every n tested, transpiled gate counts and depths fit power laws with exponents in $[1.2, 2.6]$, and qubit counts are exactly $\mathcal{O}(n) = \mathcal{O}(\log n_x) - n$ for the identity/linear circuit, $2n$ for the nonlinear one, $n+1$ for the metric.

VI. ROUTE 3: ANALOG REALIZATION IN A DEGENERATE CAVITY LASER

The third route declines the digital circuit altogether. If a physical system’s equations of motion *are* the target PDE, no embedding is required and no gate count is paid; what is paid instead is generality, since the platform solves the equation it happens to implement.

A. Platform

A digital degenerate cavity laser (DDCL) is a self-imaging $4f$ resonator supporting a large number of transverse laser modes. An aperture mask in a near-field plane defines an array of spatially separated laser sites, while an intracavity spatial light modulator in the Fourier plane programs the coupling between them. The nearest-neighbor coupling is complex and programmable in phase,

$$\kappa = K e^{i\varphi}, \quad (20)$$

with K the coupling strength and φ the coupling phase. Figure 11 shows the near-field laser array and the far-field intensity distribution that reports the array’s collective coherence.

B. From laser rate equations to Burgers

The coupled-laser dynamics obey the rate equations

$$\dot{E}_m = \frac{1}{\tau_c} \left[(G_m - \alpha_m) E_m + \sum_n \kappa_{mn} E_n \right], \quad (21)$$

together with the gain equation for G_m . Writing $E_m = A_m e^{-i\theta_m}$, the phase dynamics for complex nearest-neighbor coupling contains

$$\dot{\theta}_m = \Omega_m + \frac{K}{\tau_c} \sum_{s=\pm 1} \frac{A_{m+s}}{A_m} \sin(\theta_{m+s} - \theta_m - \varphi). \quad (22)$$

In the phase-only regime — approximately equal amplitudes $A_m \simeq A_0$ and identical natural frequencies — and after removing the common global rotation, this reduces to the nearest-neighbor Sakaguchi–Kuramoto model

$$\dot{\theta}_m = K_{\text{eff}} \sum_{s=\pm 1} \sin(\theta_{m+s} - \theta_m - \varphi), \quad K_{\text{eff}} = K/\tau_c. \quad (23)$$

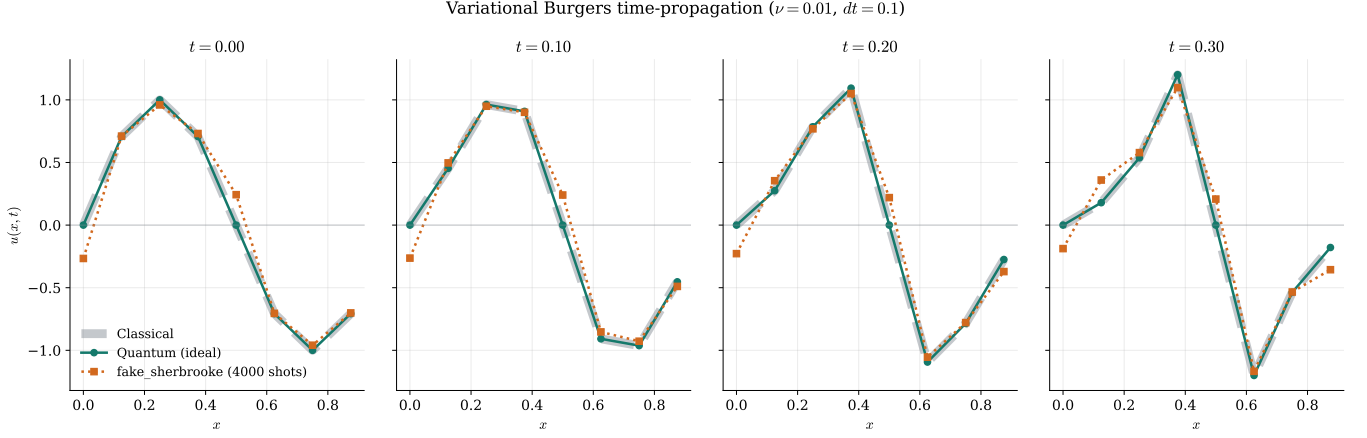


FIG. 9. Burgers propagation via Route 2: classical (dashed) versus ideal quantum (solid, matching to $\sim 10^{-6}$) versus noisy hardware readout (dotted). Viscosity here is ten times smaller than in Fig. 2, so advective steepening is far more visible before diffusion catches up, while the maximum principle of Sec. I still caps $\|u(t)\|_\infty$.

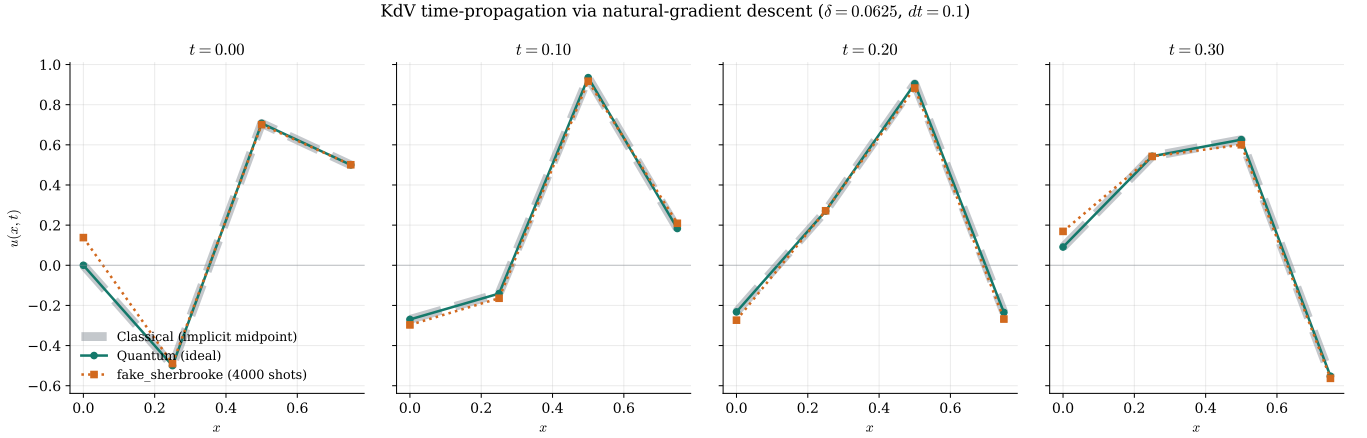


FIG. 10. KdV propagation via Route 2, with the implicit-midpoint target of Sec. VD. Dispersion replaces Burgers' diffusion: instead of damping gradients it spreads wavelengths at different phase velocities, turning would-be steepening into an oscillatory wave train rather than a shock. That mechanism is conservative, not dissipative, which is why $\|u\|_2$ is held constant to 10^{-10} classically and $\sim 10^{-7}$ by the quantum fit, where it strictly decreases for Burgers.

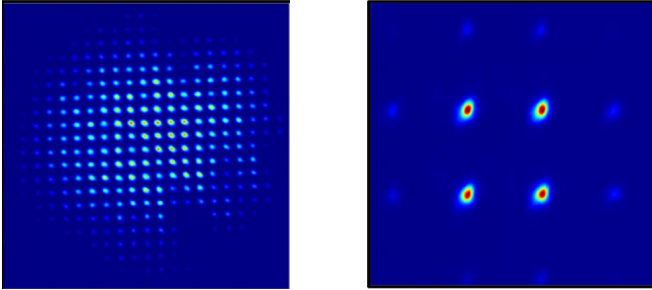


FIG. 11. The analog platform. Left: near-field plane, resolving the individual laser sites of the programmable array. Right: far-field intensity distribution, which reflects the collective coherence of the coupled array. Images reproduced from the platform description of Ref. [19].

Introducing $x = ma$ with a the site spacing, taking $\theta_m(t) \rightarrow \theta(x, t)$, and expanding for small nearest-neighbor phase differences gives

$$\theta_t = K_{\text{eff}} a^2 \cos \varphi \theta_{xx} + K_{\text{eff}} a^2 \sin \varphi (\theta_x)^2. \quad (24)$$

Defining the phase-gradient field $u = -2K_{\text{eff}} a^2 \sin \varphi \theta_x$ yields exactly the one-dimensional viscous Burgers equation,

$$u_t + u u_x = \nu_{\text{eff}} u_{xx}, \quad \nu_{\text{eff}} = K_{\text{eff}} a^2 \cos \varphi. \quad (25)$$

The physical content of Eq. (25) is worth stating plainly: the optical phase gradient plays the role of the Burgers field, and the two halves of the complex coupling split the two halves of the equation between them. The dissipative component $\cos \varphi$ produces the effective viscosity; the reactive component $\sin \varphi$ produces the advection. Tuning one experimental parameter therefore

TABLE IX. Correspondence between the Burgers model and the DDCL platform [19].

Burgers quantity	Experimental realization
x	laser-site position
$\theta(x, t)$	optical phase field
$u(x, t)$	phase-gradient field
K_{eff}	coupling strength
φ	phase of the complex coupling
a	spacing between neighboring sites
$\nu_{\text{eff}} = K_{\text{eff}} a^2 \cos \varphi$	effective viscosity

tunes the effective Reynolds number continuously — the same quantity that, in Sec. IV B, was the binding constraint on Route 1.

C. What the platform delivers, and what it does not

For a stationary state $u_t = 0$, Eq. (25) integrates once to $u_x + \tan \varphi u^2 = C$, whose bounded solution is the hyperbolic front

$$u(x) = u_0 \tanh\left(\frac{x - x_0}{\ell}\right), \quad (26)$$

with ℓ the characteristic front width and x_0 fixed by the boundary conditions. Under open boundary conditions, the experimentally reconstructed stationary phase profile therefore gives direct access to the corresponding stationary Burgers solution. Table IX lists the full correspondence.

Three limitations should be stated as clearly as the mapping. The correspondence is an effective phase-only description of the full rate equations, and requires approximately equal amplitudes, identical natural frequencies, small nearest-neighbor phase differences, and a long-wavelength continuum limit; outside those conditions the array is not solving Burgers. The accessible observable demonstrated here is the *stationary* profile, not a time-resolved trajectory, so the analog route as presented is not yet a like-for-like replacement for the digital routes' propagation results of Figs. 2 and 9. And the platform is not programmable across equations: it realizes Burgers, not KdV, and there is no analog of the truncation-order or block-encoding knobs that make Routes 1 and 2 general-purpose. What it does supply is an independent, non-quantum physical realization of the same equation, and evidence that Burgers dynamics are physically generic rather than an arbitrary benchmark.

VII. THE THREE ROUTES COMPARED

Table X places the three routes side by side. The comparison is not a ranking: they answer different questions,

and the useful content is in *where* each one's cost sits.

Read across the table, three statements follow.

a. Route 1 buys memory and nothing else, in one dimension. Its qubit count is genuinely exponentially smaller than classical memory, and no part of the analysis erodes that. Everything else is more than classical: matched ϵ scaling with large extra factors, inherently sequential depth, a state rather than a field as output, and a convergence guarantee confined to $\text{Re} < \pi/2$. The worked instance of Table VI makes the gap concrete at nine to ten orders of magnitude after error correction. A crossover in *work* requires three spatial dimensions, observable-only readout, and a polynomially bounded κ_V — and the non-normality of Carleman matrices leaves the last of these open.

b. Route 2 is cheap because it moved the hard part, not because it removed it. Its circuits are shallow and its scaling polylogarithmic, but the nonlinear work reappears as a nonconvex optimization solved afresh at every timestep with no convergence guarantee, and — in our implementation — one residual $\mathcal{O}(2^n)$ classical evaluation whenever the certificate of Proposition 3 fails to apply. We reduced that residual to ~ 55 calls per timestep and we can bound when it is needed, but making the residual fraction provably $\mathcal{O}(\text{poly}(n))$ rather than empirically small remains open. This is the honest statement of the method's status: a demonstrated present-day capability with a clearly localized theoretical gap.

c. Route 3 removes the embedding by giving up programmability. A medium whose native dynamics are the target equation pays no gate cost at all, and tunes the very parameter that obstructs Route 1 by rotating a coupling phase. The price is that it solves one equation, under conditions that must hold for the effective description to be valid, and delivers what the optics can be made to report.

Underlying all three, the discretization results of Sec. II are what make the comparison meaningful at all. Without the closure of Proposition 1, a discretized Burgers solution can violate the maximum principle that defines the problem, and a discretized KdV solution drifts by 11% in a norm the equation conserves exactly. Without the integrator result of Sec. II C, a variational propagator inherits a monotone energy injection that no circuit fidelity can remove. Neither is visible in a gate count.

VIII. CONCLUSION

We have costed three routes from the same pair of nonlinear PDEs to a quantum-encoded solution, and held all three to the same physical standard. The Carleman-QSVT route works — we built it, and every quantum result matches its classical counterpart to numerical precision — but its convergence is provable only for $\text{Re} < \pi/2$, its block encoding as usually constructed carries a subnormalization $\alpha \simeq n_x^{N_C-1} \lambda_{\text{eff}}$ that erases the nominal exponential compression, its sequential stepping requires

TABLE X. The three routes compared, at the problem sizes actually reached in this work. “Measured” quantities were built, transpiled, and run; “projected” quantities were derived and calibrated against published gate-level data. Classical reference: pseudo-spectral solve of the same instance.

	Route 1: Carleman-QSVT	Route 2: variational	Route 3: DDCL analog
Linear embedding required	yes, dimension $\sum_j n_x^j$	no	no
Nonlinearity handled by	the embedding	classical optimizer	the physical medium
Qubits (measured)	6–14	2–6	—
Gates per solve (measured)	2.6×10^6 – 1.7×10^9	29–351	—
Scaling in grid size	$\mathcal{O}(n_x^4)$ dense encoding	$\mathcal{O}(\text{poly log } n_x)$	fixed by array size
Asymptotic depth (projected)	$\tilde{\mathcal{O}}(\kappa_V T N_C^3 \text{Re}^{-1} \epsilon^{-1})$	not derived	—
Convergence guarantee	yes, but only $\text{Re} < \pi/2$	none (nonconvex)	effective-model validity
Runs on present hardware	no	yes	yes (is hardware)
Accuracy achieved	numerical precision vs. classical	2×10^{-8} – 4×10^{-6} ideal	stationary profile
Solves KdV as well	yes	yes	no

a history-state dilation to be viable at all, and its assembled depth $\tilde{\mathcal{O}}(\kappa_V T N_C^3 \text{Re}^{-1} \epsilon^{-1})$ evaluates to 1.3×10^{12} logical two-qubit gates where a laptop needs 4.4×10^6 floating-point operations. The variational route reaches the same physics with circuits of at most 351 gates on present-day hardware, at the cost of a nonconvex optimization with no convergence guarantee. The analog route reaches it with no circuit at all, at the cost of programmability.

The useful output is the decomposition, not a winner. Of Route 1’s five cost drivers, the qubit count is a clean exponential win; the block encoding and the sequential stepping are engineering problems with identifiable fixes; the readout is a hard barrier only if the full field is wanted; and exactly one — the Carleman convergence radius — is fundamental within the framework. Two methodological points generalize beyond this problem. First, state which condition number is meant: $\alpha/\sigma_{\min}$ and $\sigma_{\max}/\sigma_{\min}$ differ by a factor of 3.5 in the published Burgers instance, and only the former sets the cost. Second, report the polynomial accuracy alongside κ and the degree, without which published pairs cannot be compared.

Finally, the choice of test problem is not innocent. Burgers is linearizable in closed form by Cole–Hopf, so a Burgers demonstration establishes that circuits compile and produce the right answer — valuable, and how we use it — but cannot by itself establish that a method handles nonlinearity. Carrying KdV through every stage is what makes the comparison a test rather than an exhibition.

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DATA AND CODE AVAILABILITY

All circuits, classical reference solutions, resource-counting scripts, and the numerical validation of every formula reported here are available in the accompanying repository. The asymptotic analysis of Sec. IV is reproduced by three short Python scripts requiring only `numpy` and `scipy`; the circuit results of Secs. III and V are reproduced by four Qiskit notebooks that run end to end.

AI DISCLOSURE

The physics, algorithms, and the great majority of the code in this work are the authors’ own: the choice of Carleman linearization and QSVT, the variational natural-gradient formulation, the discretization, the error-budget analysis, and the fixes described throughout originated with the authors. Claude (Anthropic) was used interactively, under the authors’ direction and review, to help implement specific pieces of that design, debug issues the authors identified, run numerical verification, and draft parts of this manuscript’s text. All physical claims, discretization choices, and numerical results were independently verified by the authors against classical reference calculations rather than accepted on the model’s assertion alone; every proof was checked step by step; every cited reference was checked against its original source before inclusion.

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